

Diagnose what the receiver actually consumed.

A service can be reachable while vehicle inputs are stale. Make that difference observable, then prove failure and recovery with a repeatable campaign.

For operators

Locate lost receiver observations, retain fault history and inspect the evidence behind the result.

For contributors

Reuse a real receiver-to-OpenSOVD example and a narrow native storage fix with reproducible tests.

Actual native components and simulated vehicle physics. Deterministic assertions decide acceptance.

[Implementation handover](#)

[Claim and evidence map](#)

PRESERVE THE EXISTING VEHICLE FUNCTION

Control keeps its path. Diagnosis observes it.



Control return ← same bridge / managed connection ← native Cruise Control

Existing VCU normalization applies the returned throttle to the physical actor.

Receiver observation → separate OpenSOVD process

Freshness, actual state and source identity; native reporter / fault manager / persistent storage. Fault history uses labelled App data.

Bridge source and control algorithm preserved. Local GRE is verified; distributed VPN and native test executor are not claimed.

[Receiver observation patch](#)

[Detailed original architecture](#)

SHOW LIVE EXECUTION, OR EXPLICITLY SELECT THE SAVED RECORDING

Disturb. Diagnose. Recover.

1

Observe

Actual motion, accepted receiver speed, source/session identity and returned native control.

2

Interrupt

Stop the owned managed link. Accepted-data age increases; the communication fault becomes active.

3

Restore

Fresh data returns; held recovery clears the active fault while preserving history.

300 ms age threshold

100 ms debounce

150 ms held recovery

Stall diagnosis while control return continues; restart diagnosis / fault manager and inspect retained history.

The harness supplies pedal, engagement and waypoint steering. Native Cruise Control and the existing VCU supply throttle. Recovery is not a claim of automatic engagement.

[Open recorded physical replay — historical, not live](#)

[Run the documented live campaign](#)

Acknowledgement should match the selected persistence policy.

Before

Cache mutation can be visible to a manager restart in the same process while a new OS process sees no stored record.

The separate-process regression fails against the original constructor.

Prepared fix

Opt-in write-through flushes before acknowledging mutations and propagates backend errors.

Default construction keeps its existing policy. One upstream file changes.

Actual native IPC reporting, query and process-restart evidence accompanies the patch.

A failed flush can leave the in-memory mutation intact. Process-restart persistence is verified; hardware or power-loss durability is not guaranteed.

Show the evidence at its actual level.

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Core / physical assertions passed

Fixture inputs in the core run; actual CARLA motion in the physical run.
Ten conditional or deferred checks are skipped in each.

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Native / VCU / actor matches

Positive native request, matching VCU scaling and subsequent actuator throttle, with zero operator pedal.

Fresh source clones and native build directories passed controller unit tests, diagnostic tests / Clippy and both campaigns. Owned teardown and original baseline comparisons pass.

Self-run shares host, immutable images and LLVM. Human reproduction remains pending. No E2E sample identifier or actuator-latency guarantee is claimed.

[Fresh-build verification](#)

[Physical verdicts](#)

[Preservation recheck](#)

Make the integration easier to operate and extend.

Web dashboard — specified

OpenSOVD Diagnosis and Test Manager on the openDuT-managed bench: status, fault history, campaign control, cleanup and evidence.

Live UI implementation is pending. The saved replay is available now.

ThreadX + AutoSD — proposed

An actual ThreadX sensor / watchdog target and an AutoSD VM deployment environment, with interruption / recovery evidence.

Neither runtime integration nor its potential technology bonus is claimed as achieved.

AAOS/FOTA remains deferred until the user's real asset arrives. No update success is simulated.

[Dashboard specification](#)

[Bounded runtime proposal](#)

A reusable integration and a reviewable fix.

- Actual receiver diagnosis with source identity, unknown / stale states and fault history.
- Repeatable campaigns on a real openDuT-managed connection, with honest verdicts and scoped cleanup.
- A narrow fault-storage patch, failing regression, native checks and continuation packet.
- Pinned setup and recorded fallback for the next contributor.

Next acceptance: another contributor runs the documented campaign and records the result.

Prepared 4 October 2026. Event-start revision / delta must be recorded later. No maintainer approval, public PR, merge, event eligibility or production safety claim.